Part 1: Theoretical Understanding (30%)

1. Short Answer Questions

Q1: Define algorithmic bias and provide two examples of how it manifests in AI systems.
Definition: Algorithmic bias occurs when AI systems produce unfair or prejudiced outcomes due to biased training data, flawed design, or systemic inequalities embedded in the algorithms.
Examples:

- Facial recognition: Systems misidentify people of color at higher rates compared to lighter-skinned individuals, leading to discriminatory outcomes in law enforcement.
- Hiring algorithms: Recruitment AI trained on historical data may favor male candidates over female ones if past hiring practices were biased.

Q2: Explain the difference between transparency and explainability in AI. Why are both important?
Transparency: Refers to how open and clear the AI system is about its design, data sources, and decision-making processes. It's about visibility into the system's inner workings.
Explainability: Refers to the ability to interpret and understand why an AI system made a particular decision or prediction. It's about making outcomes understandable to humans.
Importance:
Transparency builds trust by showing that the system isn't hiding critical details.
Explainability ensures accountability, allowing users and regulators to challenge or correct unfair or harmful decisions.

Q3: How does GDPR (General Data Protection Regulation) impact AI development in the EU?
GDPR enforces strict data protection rules, shaping how AI systems are built and deployed:
Requires lawful, fair, and transparent data processing.
Grants individuals rights such as data access, rectification, and erasure ("right to be forgotten").
Limits automated decision-making that significantly affects individuals unless safeguards are in place.
Encourages privacy by design, meaning AI developers must integrate data protection principles from the start.

2. Ethical Principles Matching

A) Justice: Fair distribution of AI benefits and risks.
B) Non-maleficence: Ensuring AI does not harm individuals or society.
C) Autonomy: Respecting users’ right to control their data and decisions.
D) Sustainability: Designing AI to be environmentally friendly.